MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

DEPARTMENT OF CIVIL ENGINEERING			
END SEMESTER EXAMINATION		**NOVEMBER, 2024**	
Level	**B.Tech**	**Curriculum**	**2022**
Program Name	**B. TECH**	**Course Type**	**C**
Course Name	**WATER SUPPLY ENGINEERING**	**Course Code**	**CIE 2125**
Semester	**III**	**Max Marks**	**50**
Date	**27-11-2024**	**Duration**	**3 hours**

Instructions:
- **Answer ALL the questions.**
- **Data not provided may be suitably assumed**

Q. No.	Question	Marks	NBA		BL
			CO	PO	
1A	Explain the concept of water demand. Describe the factors contributing to its variations and summarize the relationships between average daily demand, maximum daily demand, and maximum hourly demand.	**4**	CO1	1,2,6 ,7,12	**L2**
1B	Identify and explain the factors that influence per capita water demand.	**4**	CO1	1,2,6 ,7,12	**L2**
1C	Describe the impact of excess Nitrates and Hardness in water.	**2**	CO2	1,6,7 ,8,12	**L2**
2A	Explain in detail the theory of Filtration.	**4**	CO3	1,2,3 ,7,12	**L2**
2B	Illustrate the Zeolite process used for the removal of hardness.	**4**	CO3	1,2,3 ,7,12	**L2**
2C	Briefly explain any two types of chlorination.	**2**	CO3	1,2,3 ,7,12	**L2**
3A	A coagulation - sedimentation plant clarifies 60 MLD of water. The quantity of Alum required at the plant is 30 mg/l. If the raw water is having an alkalinity equivalent to 11 mg/l of $CaCO_3$, determine the quantity of Alum and quick lime (containing 85% of CaO) required per year by the plant. Given the molecular weights as: Al = 27, S = 32, O = 16, H = 1, Ca= 40 and C =12. Assume the following: 1 mole of Alum = 3 moles of $Ca(HCO_3)_2$ = 1 mole of $CaCO_3$.	**4**	CO4	1,2,3 ,7,8, 12	**L3**

<table>
<tr><td>3B</td><td>A sample of raw water contains 550 mg/l alkalinity, 200 mg/l hardness as $CaCl_2$ and 90 mg/l hardness as $MgCl_2$. Compute the quantities of pure lime and soda required to treat 1,80,000 litres of water. If slaked lime (92% pure) is available in place of pure lime, what will be the required quantity of slaked lime.
Use the following relation:
a) Total Lime requirement = 0.74*(2*temperorary Magnesium hardness + temperorary Calcium hardness + permanent Magnesium hardness)
b) Total Soda requirement = 1.06*(permanent Calcium hardness + permanent Magnesium hardness)</td><td>4</td><td>CO4</td><td>1,2,3 ,7,8, 12</td><td>L3</td></tr>
<tr><td>3C</td><td>Determine the dosage and chlorine demand at a water treatment plant with a flow of 30,000 m^3/day, having a daily chlorine requirement of 9 kg/day. The residual chlorine after a 10-minute contact time is 0.18 mg/L.</td><td>2</td><td>CO4</td><td>1,2,3 ,7,8, 12</td><td>L3</td></tr>
<tr><td>4A</td><td>Determine the economical diameter of a rising main for pumping 13,00,000 liters per hour of water over a distance of 3,500 m from a canal (elevation 30 m) to storage reservoir (elevation 90 m). Also calculate the water hourse power of pump. Assume f = 0.01.</td><td>5</td><td>CO5</td><td>1,2,6 ,7,12</td><td>L3</td></tr>
<tr><td>4B</td><td>Illustrate with a neat sketch the house service connection.</td><td>3</td><td>CO5</td><td>1,2,6 ,7,12</td><td>L2</td></tr>
<tr><td>4C</td><td>Explain the methods for leak detection in water supply system.</td><td>2</td><td>CO5</td><td>1,2,6 ,7,12</td><td>L2</td></tr>
<tr><td>5A</td><td>Using analytical method calculate the storage required to supply the demand shown in the following table. The pumping is to be carried out at a constant rate from 8am to 2pm.
<table>
<tr><td>Time (hours)</td><td>0 - 4</td><td>4 - 8</td><td>8 - 12</td><td>12 - 16</td><td>16- 20</td><td>20 -24</td></tr>
<tr><td>Demand (MLD)</td><td>0.48</td><td>0.87</td><td>1.33</td><td>1.0</td><td>0.82</td><td>0.54</td></tr>
</table></td><td>5</td><td>CO5</td><td>1,2,6 ,7,12</td><td>L4</td></tr>
<tr><td>5B</td><td>List various pipe appertenences used in water supply system. Explain the working of air relief value.</td><td>3</td><td>CO5</td><td>1,2,6 ,7,12</td><td>L2</td></tr>
<tr><td>5C</td><td>What is the significance of fire hydrants? Explain Post type fire hydrant in detail</td><td>2</td><td>CO5</td><td>1,2,6 ,7,12</td><td>L2</td></tr>
</table>

MARK DISTRIBUTION (CO-wise & BTL-wise)

CO / CLO	Total Marks
1	***08***
2	***02***
3	***10***
4	***10***
5	***20***

BL	Total Marks
2	30
3	15
4	5
5	-